PROFESSIONAL PAPERS OF THE CORPS OF ENGINEERS, U. S. ARMY.

No. 24.

REPORT

UPON

THE PRIMARY TRIANGULATION

OF THE

UNITED STATES LAKE SURVEY.

BY

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AIDED BY THE

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WASHINGTON:
GOVERNMENT PRINTING OFFICE.
1882.

Differences of personal equation between Assistant O. B. Wheeler and General Comstock, and also between Assistant G. Y. Wisner and General Comstock, were determined. The method followed was for one observer to note the transit of a star over the first tally of five wires, and the other to note the transit over the last tally of five wires, the observers alternating in tallies used. The following results were obtained. The minus sign prefixed to a value for personal equation signifies that the first-named observer observed the transit of a star earlier than the second:

Date.	Observers.	Personal equation.
1871.		*s.* *s.*
July 7	C. B. C.—O. B. W.....	—0. 007±0. 015
9	.....do	—0. 098±0. 032
Aug. 3	.....do	—0. 071±0. 010
4	.....do	—0. 056±0. 012
June 21	C. B. C.— G. Y. W.....	—0. 091±0. 029
22	..... do	—0. 181±0. 022
24	.....do	+0. 005±0. 026

In the Lake-Survey Report for 1872 mean results have been obtained by giving to the result of each night's work a weight depending on the probable error of that night's work. But as the discrepancies between results from different nights far exceed the probable errors derived from that night's work, it must be concluded that the personal equation varied from night to night, and the mean of the different night's results, attributing equal weights to each, will be a value more nearly correct, and a better value of the probable accuracy of the result will be obtained by considering the discrepancies between the results of each night's work and the mean of all nights' work than from attributing to each night's work the accuracy which results from its own work alone. Taking, then, the arithmetical mean of the results of different nights and computing its probable error from the discrepancies between this mean and the separate nights' results, we have for adopted personal equations,

$$\text{C. B. C.}-\text{O. B. W.} = -0.058^{s} \pm 0.013^{s}$$
$$\text{C. B. C.}-\text{G. Y. W.} = -0.089^{s} \pm 0.034^{s}.$$

From these values there results—

$$\text{G. Y. W.}-\text{O. B. W.} = +0.031^{s} \pm 0.036^{s}.$$

Applying for June 15 the personal equation between C. B. C. and O. B. W., and for June 26 and July 11 that between G. Y. W. and O. B. W., there results for difference of longitude of the two instruments,

	m.	*s.*
June 15, 1871	36	05.90
June 26, 1871	36	05.80
July 11, 1871	36	05.76

The mean of these values is $36^{m}\ 05^{s}.82$.

The probable error of this mean, if derived from the discrepancies between it and the separate results, would be less than $\pm 0^{s}.03$, but as the probable error in the personal equation of Assistant O. B. Wheeler and Assistant Wisner reaches $\pm 0^{s}.036$, the probable error in the result may reach $\pm 0^{s}.05$, giving $36^{m}\ 05^{s}.82 \pm 0^{s}.05$.

At Duluth the instrument was $0^{s}.143$ east of primary station North Base, being east 150.1 feet and north 110.2 feet from that station; so that we have station North Base west of east transit pier of Lake-Survey Observatory, built in 1871,

$$\mathbf{36^{m}\ 05^{s}.96 \pm 0^{s}.05.}$$

The details of this work can be found in Assistant O. B. Wheeler's report of the reduction, published in the Lake-Survey Report for 1872.

LONGITUDE OF STATION FORT HOWARD.

§ **3**. A telegraphic determination of the difference of longitude between Fort Howard, Wisconsin, and Detroit was made in 1864, on the nights of August 8, 10, and 13. The observer at

Detroit on the first two nights was Colonel W. F. Raynolds, and on the last night Assistant S. W. Robinson. The observations at Fort Howard were made by Professor C. A. Young. At Detroit the Troughton & Simms transit of 43-inch focal length and 3-inch objective, sidereal clock No. 184 and chronograph No. 216 by Bond & Sons were used. At Fort Howard Würdemann transit No. 15 of 31-inch focal length and 2½-inch objective, with Bond & Sons sidereal clock No. 256, and chronograph No. 245 were used. The telegraphic circuit was continuous between the two stations, the distance being 526 miles.

The plan of observation followed was that of recording the transits of the same stars at the two stations on both chronographs. In the reduction, the differences of the times of transit of the same stars at the two stations, corrected for instrumental errors and rate of clock, were computed from the Detroit chronographic record. These differences, when corrected for personal equation and wave and armature time, will give the difference of longitude. The following table gives these differences for each night and each star, with their means for the separate nights. The probable errors are deduced from the discrepancies between the means and the individual results.

Results for difference of longitude between Detroit and Fort Howard.

[Uncorrected for personal equation and wave and armature time.]

Stars observed.	August 8.		August 10.		August 13.	
	Individual results.	Residuals.	Individual results.	Residuals.	Individual results.	Residuals.
	m. *s.*	*s.*	*m.* *s.*	*s.*	*m.* *s.*	*s.*
α Vulpeculæ	.....	.....	.....	.....	19 52.057	0.447
β Sagittæ	19 51.598	0.017	.....	.....	51.970	.360
β Aquilæ	51.526	.055	.....	.....	51.837	.227
α^2 Capricorni	51.487	.094	19 51.752	0.054	51.537	.073
ρ Capricorni	51.345	.236	51.647	.051	51.566	.044
α Delphini	51.287	.294	51.673	.025	51.397	.213
μ Aquarii	51.493	.088	51.883	.185	.....	.....
12 Aquarii	51.584	.003	51.681	.017	51.588	.022
ζ Cygni	51.248	.333	51.552	.146	51.586	.024
ζ Capricorni	.....	.....	51.470	.228	51.643	.033
ξ Aquarii	51.796	.215	51.823	.125	51.838	.228
14 Pegasi	51.331	.250	.....	.....	51.126	.516
α Aquarii	51.560	.021	51.681	.017	51.470	.140
θ Aquarii	51.330	.251	.....	.....	51.587	.023
ζ Aquarii	51.441	.140	51.773	.075	.....	.....
ζ Pegasi	51.671	.090	51.767	.069	51.605	.005
λ Aquarii	51.671	.090	51.749	.051	51.432	.178
α Pegasi	51.649	.068	51.550	.148	51.332	.278
γ Piscium	51.820	.239	51.593	.105	51.591	.019
κ^1 Piscium	51.891	.310	51.941	.243	51.955	.345
ι Piscium	51.831	.250	51.806	.108	51.549	.061
δ Sculptoris	51.115	.466	51.655	.043	51.378	.232
29 Piscium	51.753	.172	.....	.....	51.305	.305
γ Pegasi	.....	.....	51.687	.011	.....	.....
10 Ceti	.....	.....	51.765	.067	51.440	.170
δ Andromedæ	51.550	.031	51.536	.162	51.411	.199
ϕ^2 Ceti	51.739	.158	51.757	.059	51.315	.295
ε Piscium	51.827	.246	51.723	.025	51.578	.032
θ Ceti	51.583	.002	51.455	.243	51.302	.308
η Piscium	.....	.....	51.829	.131	51.770	.160
ν Piscium	.....	.....	.....	.....	51.918	.308
β Arietis	51.828	.247	.....	.....	.....	.....
α Arietis	51.722	.141	.....	.....	.....	.....
Means and probable errors	19 51.581 ± 0.227		19 51.698 ± 0.017		19 51.610 ± 0.030	

$19^m\ 51^s.0$ and for x_1, x_2, x_3, and x_4, their observed values, there result the following equations:

$$(y) \quad -(x_1)=+0^s.839+\Delta_1, \text{ with weight } p_1=\left(\frac{1}{0.027}\right)^2$$

$$(y) \quad -(x_1)=+0^s.956+\Delta_2, \text{ with weight } p_2=\left(\frac{1}{0.017}\right)^2$$

$$(y)+(x_2)+(x_3)=+0^s.632+\Delta_3, \text{ with weight } p_3=\left(\frac{1}{0.030}\right)^2$$

$$(x_1)=\ 0 \quad +\Delta_4, \text{ with weight } p_4=\left(\frac{1}{0.022}\right)^2$$

$$(x_2)=\ 0 \quad +\Delta_5, \text{ with weight } p_5=\left(\frac{1}{0.024}\right)^2$$

$$(x_3)=\ 0 \quad +\Delta_6, \text{ with weight } p_6=\left(\frac{1}{0.015}\right)^2$$

$$(x_4)=\ 0 \quad +\Delta_7, \text{ with weight } p_7=\left(\frac{1}{0.016}\right)^2$$

In addition we must have—

$$x_1+(x_1)+x_2+(x_2)+x_3+(x_3)=x_4+(x_4)$$

or

$$(x_1)+(x_2)+(x_3)-(x_4)=-0^s.143$$

Substituting the value of (x_4) from this last equation in the seventh of the above equations and solving so as to make $[p\Delta^2]$ a minimum, there results

$$(y)=+0^s.840\pm0^s.059,$$

and hence

$$y+(y)=+19^m\ 51^s.840\pm0^s.059.$$

The wave and armature time was computed from the differences between the two clock times of transit of those stars whose transits at the two stations were recorded on both chronographs. The difference between such differences for any star gives double the wave and armature time. The results obtained are given in the table below, the separate values being weighted inversely as the squares of their probable errors, which latter are derived from the discrepancies between the individual values and their mean for any date.

Table of results for wave and armature time.

Date.	Wave and armature time.	Probable error.	No. of stars observed.
1864.	*s.*	*s.*	
August 8	0.0462	±0.0017	29
10	0.0434	±0.0017	19
13	0.0461	±0.0020	15
Weighted mean	0.045	±0.001	

Subtracting the mean value of the wave and armature time from the above value of $y+(y)$, there results for the difference of longitude between the transit-post at Detroit and the transit-post at Fort Howard, $19^m\ 51^s.80\pm0^s.06$.

The Detroit transit-post of 1864 was $0^s.127$ east of the east pier of the present Lake-Survey Observatory, and the post at Fort Howard was 6910.8 feet or $6^s.360$ east, and north 4884.2 feet from Fort Howard triangulation station. Applying these corrections, we have Fort Howard station west of east pier of Lake-Survey Observatory at Detroit,

$$\mathbf{0^h\ 19^m\ 58^s.03\pm0^s.06.}$$

§ **5.** For convenience of reference, the results for astronomical latitudes and longitudes given in this chapter and in Chapters XXIII and XXV will be collected in a single table arranged in the order of latitudes. All longitudes depend on the longitude of the east pier of the Lake-Survey Observatory of 1871 at Detroit, given in Chapter XXV, § 1.

Table of Astronomical Latitudes and Longitudes.

Station.	Latitude.	Longitude.	Details given on page—	Point of reference.
	° ′ ″	° ′ ″		
St. Ignace, Ontario	48 47 28.65		622	Triangulation station.
Tip Top, Ontario	48 16 25.78		733	Do.
Isle Royale, Mich	48 07 55.22		733	Do.
Farquhar's Knob, Minn	47 52 35.27		733	Do.
Michipicoten, Ontario	47 45 20.58		733	Do.
Gargantua, Ontario	47 34 38.80		734	Do.
Copper Harbor (Fort Wilkins), Mich	47 28 03.15		734	Stone post marked S.
Vulcan, Mich	47 26 44.58		625	Triangulation station.
Mount Houghton, Mich	47 24 15.83		734	Do.
Sawteeth East, Minn	47 23 09.00		734	Do.
Wheal Kate, Mich	47 04 18.12		734	Do.
Outer Island, Wis	47 04 14.51		735	Do.
Portage Entry, Mich	46 58 47.72		735	Stone post marked S.
Crebassa, Mich	46 58 41.21		735	Light-house.
Buchanan, Minn	46 56 24.45		735	Triangulation station.
Huron Mountains, Mich	46 52 53.07		626	Do.
South Base, Keweenaw Point, Mich	46 52 22.35		623	Do.
Porcupine Mountains, Mich	46 47 03.80		735	Do.
Whitefish Point, Mich	46 46 17.62		736	Light-house.
North Base, Minnesota Point, Minn	46 45 28.32	92 04 33.00	625, 717	Triangulation station.
Brulé River, Wis	46 45 20.11		736	Do.
South Base, Minnesota Point, Minn	46 42 51.42		736	Do.
Aminicon, Wis	46 41 36.32		736	Do.
Grand Island, Mich	46 33 39.77		736	Light-house.
Thone's Hill, Mich		87 26 49.95	723	Stone post.
Marquette, Mich	46 32 54.53		737	Light-house.
Sault Ste. Marie, Mich	46 30 10.29		737	Do.
Iroquois Point, Mich	46 29 04.84		737	Do.
Escanaba, Mich	45 44 35.05		737	Do.
Ford River, Mich	45 41 05.34	87 06 01.35	626, 722	Triangulation station.
Burnt Bluff, Mich	45 41 03.88		737	Do.
Beaver Island, Mich	45 34 28.78		738	Light-house.
Cedar River, Mich	45 25 43.13		738	Triangulation station.
Rock Island, Wis	45 25 35.49		738	Light-house.
Boyer's Bluff, Wis	45 25 09.30		738	Triangulation station.
Door Bluff, Wis	45 17 45.69		738	Do.
Menomonee, Mich. (near Marinette, Wis.)	45 05 12.78	87 35 27.75	739, 753	Do.
Green Island, Wis	45 03 24.80		739	Light-house.
Thunder Bay Island, Mich	45 02 17.27		739	Do.
Saint Paul, Minn	44 56 42.96	93 05 34.05	739, 753	Custom-house.
Ogdensburg, N. Y	44 41 53.19	75 30 18.90	739, 753	Light-house.
Clay Banks, Door County, Wis	44 41 19.45		740	Azimuth station.
Frankfort, Mich	44 38 53.06		740	Do.
Red Wing, Minn	44 33 44.64	92 31 49.65	740, 760	Spire of Catholic church.
Fort Howard, Wis	44 30 30.28	88 02 34.05	627, 718	Triangulation station.
Kewaunee, Wis	44 25 41.12		740	Azimuth station.
Deadwood, Dak		103 43 18.6	766	Post-office.
Manistee, Mich	44 16 45.53		741	Azimuth station.
Valley Junction, Wis	44 14 57.24		741	Southwest corner of section 31, township 21 north, range 3 east.
Valley Junction, Wis		90 25 56.70	760	Northeast corner of section 1, township 18 north, range 1 west.

Table of Geographical Positions of Secondary Points—Continued.

State and county.	Description of point.	Latitude.	Longitude.
MICHIGAN—Continued.		° ′ ″	° ′ ″
Monroe County	Corner of sections 25, 26, 35, 36, township 6 south, range 6 east.	41 55 28.2	83 40 18.7
	Northeast corner of section 5, township 8 south, range 7 east.	41 49 19.8	83 36 32.4
	Southwest corner of private claim No. 432	41 54 42.6	83 28 29.9
	Spire of Methodist Episcopal church on southeast corner of Monroe and Second streets, Monroe.	41 54 52.4	83 23 56.6
	Spire of Presbyterian church, Monroe	41 54 53.4	83 23 51.4
	Spire of German Lutheran church, southeast corner of Scott and Third streets, Monroe.	41 54 43.4	83 23 49.1
	Light-house, Monroe	41 53 27.7	83 19 53.2
	Pointe Mouillée	42 00 49.6	83 10 53.7
Lapeer County	Southwest corner of southeast quarter of section 32, township 8 north, range 10 east.	*43 03 26.3	*83 18 55.6
	Intersection of axes of Calhoun and Liberty streets, Lapeer.	*43 03 08.6	*83 18 53.5
	Intersection of axes of Calhoun and Franklin streets, Lapeer.	*43 03 07.3	*83 18 53.4
Oakland County	Cupola of court-house at Pontiac	*42 38 22.0	*83 17 30.0
	Intersection of axes of Saginaw and Church streets, Pontiac.	*42 38 06.7	*83 17 22.6
	Spire of Methodist Episcopal church on southeast corner of Thomas and Saginaw streets, Pontiac.	*42 38 03.2	*83 17 20.5
	Southeast corner of section 29, township 3 north, range 10 east.	*42 38 07.0	83 17 15.5
Wayne County	North Wyandotte	42 12 49.2	83 08 48.5
	Sugar Island	42 05 25.4	83 08 35.4
	Barker	42 07 39.5	83 08 26.3
	Biddle	42 08 02.9	83 08 17.8
	Brodhead	42 09 30.5	83 08 15.9
	Ecorse	42 15 21.4	83 08 09.5
	River Rouge	42 16 15.4	83 07 34.9
	Field	42 17 34.2	83 06 23.1
	Astronomical post (east pier) in Lake-Survey Observatory on south side of Grand River avenue, between Park Place and Washington avenue, Detroit.	42 20 00.8	83 03 07.4
	Intersection of axes of Washington and Grand River avenues, Detroit.	42 20 01.4	83 03 04.5
	Spire of Saint Paul's Episcopal church, on northeast corner of Congress and Shelby streets, Detroit.	42 19 45.9	83 02 53.6
	Tower of fire-engine house on Russell street, Detroit	42 20 41.0	80 02 21.5
INDIANA.			
Lake County	Southeast corner of northeast quarter of section 6, township 36 north, range 7 west.	41 36 05.1	87 15 40.5
La Porte County	Light-house, Michigan City	41 43 21.1	86 54 21.8
	Spire of Deutsche Ver. Ev. Luth. St. Joh. Kirche, southwest corner of Franklin and Ninth streets, Michigan City.	41 42 45.4	86 53 59.6
	Spire of high school between Eighth, Cedar, Ninth, and Pine streets, Michigan City.	41 42 51.8	86 53 47.6
	Northwest corner of section 10, township 36 north, range 4 west.	41 35 32.8	86 52 39.0
	Southeast corner of section 10, township 37 north, range 3 west.	41 39 52.2	86 44 19.7
	Southwest corner of southeast quarter of southwest quarter of section 32, township 38 north, range 2 west.	41 41 39.0	86 40 33.4
Saint Joseph County	Spire of Christian church at New Carlisle	41 42 22.3	86 30 41.9
	Spire of Methodist Episcopal church at New Carlisle	41 42 25.7	86 30 40.8
	Corner of sections 2, 3, 10, 11, township 37 north, range 1 west.	41 40 46.4	86 30 23.6

* Astronomical.

In the following table the latitudes derived geodetically from Fort Howard, Wis., as an origin, together with their observed values, and the resulting excesses of computed over observed values, are given for many stations. The geodetic latitudes are taken from Chapter XXVII, and the observed from Chapters XXIII and XXVI. Since when the plumb-bob is deflected to the north the observed zenith moves to the south, thus diminishing the observed latitude, a plus sign before these excesses corresponds to a northerly deflection of the plumb-bob.

Computed and Observed Latitudes of Primary Stations.

Stations.	Latitudes.		Computed *minus* observed values.	Point of reference.
	Computed from latitude of Fort Howard.	Observed.		
	° ′ ″	″	″	
Saint Ignace	48 47 18.90	28.65	− 9.75	Geodetic point.
Tip Top	48 16 24.64	25.78	− 1.14	Do.
Isle Royale East	48 07 43.58	55.22	−11.64	Do.
Farquhar's Knob	47 52 40.18	35.27	+ 4.91	Do.
Michipicoten	47 45 21.66	20.58	+ 1.08	Do.
Gargantua	47 34 42.72	38.80	+ 3.92	Do.
Vulcan	47 26 46.65	44.58	+ 2.07	Do.
Mount Houghton	47 24 26.00	15.83	+10.17	Do.
Sawteeth East	47 23 19.02	09.00	+10.02	Do.
Wheal Kate	47 04 20.27	18.12	+ 2.15	Do.
Outer Island	47 04 17.09	14.51	+ 2.58	Do.
Crebassa	46 58 41.07	41.21	− 0.14	Do.
Buchanan	46 56 28.07	24.45	+ 3.62	Do.
Huron Mountains	46 52 42.30	53.07	−10.77	Do.
South Base, Keweenaw	46 52 17.76	22.34	− 4.58	Do.
Porcupine Mountains	46 47 01 05	03.80	− 2.75	Do.
North Base, Minnesota Point	46 45 27.89	28.32	− 0.43	Do.
Brulé River	46 45 17.89	20.11	− 2.22	Do.
South Base, Minnesota Point	46 42 49.44	51.50	− 2.06	Do.
Aminicon	46 41 32.15	36.32	− 4.17	Do.
Thone's Hill	46 32 47.80	54.53	− 6.73	Transit-post.
Escanaba	45 44 41.24	35.05	+ 6.19	Light-house.
Ford River	45 41 12.16	05.34	+ 6.82	Geodetic point.
Burnt Bluff	45 41 09.73	03.88	+ 5.85	Do.
Beaver Island	45 34 32.45	28.78	+ 3.67	Light-house.
Cedar River	45 25 48.61	43.13	+ 5.48	Geodetic point.
Boyer's Bluff	45 25 12.74	09.30	+ 3.44	Do.
Door Bluff	45 17 46.97	45.69	+ 1.28	Do.
Menomonee	45 05 12.76	12.78	− 0.02	Do.
Fort Howard	Initial latitude.	44 30 30.28	.....	Do.
Watertown	43 58 32.20	32.88	− 0.68	Court-house.
North Base, Sandy Creek	43 40 43.61	41.52	+ 2.09	Geodetic point.
Minnesota Junction	43 28 28.45	31.82	− 3.37	Do.
Oswego	43 26 37.28	39.57	− 2.29	Do.
Rochester	43 09 18.11	22.44	− 4.33	Court-house.
Tonawanda	43 00 3.84	07.83	− 3.99	Geodetic point.
Buffalo	42 52 41.09	44.66	− 3.57	Intersection of streets.
Dunkirk	42 29 37.84	44.30	− 6.46	New light-house of 1876.
Detroit	42 20 01.18	19 59.04	+ 2.14	Intersection of Grand River and Washington avenues.
Erie	42 09 11.85	17.49	− 5.64	Beacon light.
Monroe	41 54 52.60	48.65	+ 3.95	Court-house dome.
Chicago	41 53 20.52	22.49	− 1.97	Light-house.
Willow Springs	41 43 36.93	38.63	− 1.70	Geodetic point.
Toledo	41 39 04.54	03.65	+ 0.89	Post of 1881.
Cleveland	41 30 01.63	06.13	− 4.50	Light-house on bank.
Sandusky City	41 29 02.32	04.59	− 2.27	Geodetic point, West Base.
Fairmount	40 01 35.84	36.70	− 0.86	Geodetic point.
West Base, Olney	38 51 38.76	41.23	− 2.47	Do.
Parkersburg	38 34 51.73	53.20	− 1.47	Do.

TABLE I.—*Excesses of computed azimuths and longitudes over observed values.*

[Azimuths are computed from the initial azimuth of the line Bruce — Long Tail Point Light. Longitudes are computed from the initial longitude of Fort Howard.]

Station.	Observed latitude.	Station whose azimuth is observed.	Longitudes.			Azimuths.		
			Observed.	Computed.	Discrepancy.	Observed.	Computed.	Discrepancy.
	° ′ ″		° ′ ″	″	″	° ′ ″	″	″
North Base, Minnesota Point.	46 45 28.32	To South Base	92 04 33.00	43.26	+10.26	323 52 24.32	13.75	−10.58
Aminicon		To Lester				153 36 66.15	54.59	−11.56
South Base, Keweenaw Point.	46 52 22.35	To North Base				199 10 67.29	57.68	− 9.61
Ford River	45 41 05.34	To Cedar River	87 06 01.35	09.40	+ 8.05	31 34 60.28	53.74	− 6.54
Bruce		To Long Tail Point Light.				139 16 04.90	Initial azimuth.	
Fort Howard	44 30 30.28		88 02 34.05	Initial longitude.				
Minnesota Junction	43 28 31.82	To Horicon				272 30 45.03	47.23	+ 2.20
Willow Springs	41 43 38.63	To Shot Tower	87 51 06.75	06.36	− 0.39	224 29 41.24	39.26	− 1.98
Parkersburg	38 34 53.20	To Denver	88 01 48.96	49.66	+ 0.70	143 16 15.44	14.68	− 0.76
Toledo, longitude-post of 1881.	41 39 03.65		83 32 30.60	31.85	+ 1.25			
West Base, Sandusky	41 29 04.59	To East Base				319 36 38.60	41.20	+ 2.60
Tonawanda	43 00 07.83	To Buffalo Plains	78 53 21.16	21.91	+ 0.75	314 22 40.61	39.18	−1.43
North Base, Sandy Creek	43 40 41.52	To South Base				357 14 26.95	28.81	+ 1.86
Mannsville			76 03 26.74	14.61	−12.13			

In the following table are given some longitudes also obtained telegraphically, but in which either personal equation was not eliminated, or for other reasons the precision is less than for those given in the preceding table. They are derived from Chapter XXVI. Detroit, however, was well determined, and was the base for all the other longitude stations, but it is connected with the main triangulation in Lake Erie by a chain of small triangles running down the Detroit River, so that its geodetic position with reference to Fort Howard is less accurately known than that of a station of the primary triangulation.

TABLE II.—*Comparison of computed and observed longitudes.*

Stations.	Longitudes computed from longitude of Fort Howard.	Observed values.	Computed *minus* observed values.	Point of comparison.
	° ′ ″	″	″	
Chicago	87 36 52.20	48.90	+ 3.30	Light-house.
Menomonee	87 35 34.60	*27.75	+ 6.85	Geodetic point, Menomonee.
Thone's Hill	87 29 21.83	*07.50	+14.33	Geodetic point, Triloba.
Monroe	83 23 49.77	46.35	+ 3.42	Court-house.
Detroit	83 03 07.42	03.60	+ 3.83	Observatory-post east.
Sandusky	82 40 58.43	41 02.67	− 4.24	Geodetic point, West Base.
Cleveland	81 42 09.58	13.20	− 3.62	Light-house.
Erie	80 04 40.21	46.95	− 6.74	Beacon light.
Dunkirk	79 21 15.81	25.80	− 9.99	Light-house, 1876.
Buffalo	78 52 14.28	*18.00	− 3.72	Intersection of streets.
Rochester	77 36 50.69	51.00	− 0.31	Court-house.
Oswego	76 30 50.40	53.55	− 3.15	Geodetic point, Oswego.
Watertown	75 55 54.02	56 01.95	− 7.93	Court-house.

* Personal equation applied or eliminated.

An examination of Table I shows large discrepancies between the computed and observed longitudes at North Base, Minnesota Point, and at Mannsville.

At Minnesota Point the longitude station is on a low sand point at the west end of Lake Superior. To the east-northeast the lake deepens to 300 feet within twenty miles, while to the west,

APPENDIX III.

DIFFERENCE OF LONGITUDE BETWEEN DETROIT, MICH., AND CAMBRIDGE, MASS.

REPORT OF A. R. FLINT AND O. B. WHEELER, *Assistant Engineers.*

OFFICE UNITED STATES LAKE SURVEY,
Detroit, Mich., June 30, 1882.

SIR: In compliance with your instructions, we have the honor to make the following report on our recent determination of the difference of longitude between Detroit, Mich., and Cambridge, Mass.

The field-work was done in obedience to the following order:

OFFICE UNITED STATES LAKE SURVEY,
Detroit, Mich., April 22, 1881.

Assistant Engineer A. R. FLINT,
Detroit, Mich.:

SIR: On May 2 the determination of the difference of longitude of Detroit and Cambridge Observatory * * * will be undertaken.

Time-determinations will be made on five nights with yourself at Detroit and Assistant Engineer O. B. Wheeler at Cambridge, using chronographs and instruments of the same construction and size; the observers will then change places, and five more nights be obtained. * * *

Star-places will be taken from 539 Sterne des Fundamental-Catalogs, Berlin, 1881. Eleven wires will be observed on for time-stars, and on slow stars at least seven wires before and after reversal.

Programme for time-determination.

Level readings.
Circumpolar star reversed on.
Level readings.
Five or more well-determined time-stars.
Level readings.
Reversal.
Level readings.
Five or more well-determined time-stars.
Level readings.
Circumpolar star reversed on.
Level readings.

Then will follow the exchange of clock-signals, which will be sent alternately for $1^m\ 20^s$ from each station till two sets have been sent from each station. The observers will carefully adjust the tensions of their relays and request the repeating stations to do the same before sending clock-signals. After exchange of clock-signals another time-determination according to the above programme will be made. Every care will be taken to secure most accurate work.

Very respectfully,

C. B. COMSTOCK,
Major of Engineers and Brevet Brigadier-General.

The instructions were followed as strictly as the weather and other circumstances would permit.

The two sets of transits, clocks, chronographs, observing-keys, and other electrical apparatus were as nearly alike as possible. The observer used the same transit and observing-key at both stations, in order to eliminate any errors peculiar to their construction.

The transits, made by Würdemann, and numbered 1 and 15, respectively, were of the same size and similarly mounted. They are of about 31 inches focal length, 2½ inches clear aperture of object-glass, are provided with diagonal eye pieces and reticules of nineteen threads, and magnify about sixty-five diameters. Mr. Flint used transit No. 1 and Mr. Wheeler used No. 15. The pivot-corrections and level-values were redetermined for each transit. The values of one division of the levels were tested by means of the level trier, with the tubes in their holders, and were found to be the same as formerly used. The wire-intervals for transit No. 1 were redetermined, but not differing materially from the old values, the latter were used. The wire-intervals for a new set of wires in transit No. 15 were determined by at least twenty observations on slow stars for each wire.

Bond & Sons sidereal clock No. 256 and Bond & Sons spring-governor chronograph No. 216 were used at Detroit. Frodsham sidereal clock No. 1327, belonging to Harvard College Observatory, and Bond & Sons spring-governor chronograph No. 245 were used at Cambridge. This clock is the same as was used by the United States Coast Survey in the transatlantic longitude work of 1872, and was then designated the South Clock. The connection with the clock was through a local circuit with a battery of the same number of cups as was in use on the local circuit of the Observatory. The zinc poles of the two circuits were placed toward each other in order to prevent interference of the circuits with each other. The use of this clock was kindly granted to the Lake Survey by Professor E. C. Pickering, Director of the Observatory, and we may here state that every facility was rendered by the Director and his assistants in the execution of our work at that station.

At Cambridge the observations were made on the south pier of the United States Coast-Survey Observatory used in the transatlantic work of 1872. This station is on the grounds of the Harvard College Observatory, 108 feet, or 0ˢ.096 of time, west of the center of the great equatorial dome of the main building (see United States Coast-Survey Report for 1874, p. 164). The two granite piers there described were spanned by a marble slab 3 feet in length and 3 inches in thickness to accommodate our transits. At Detroit the observations were made on the west stone pier of the Lake-Survey Observatory (1871–'82). This pier is 5.1 feet, or in time 0ˢ.004, west of the east pier which was used in the Washington longitude work of 1871.

Usually a greater number of stars was observed than was required by the programme, and in selecting for reduction preference was given to those stars observed at both stations on the same night.

The signals were made automatically, the clock breaking the circuit every second, except the sixtieth or zero second. The clock in a local circuit repeated to the main line, and the signals were again repeated to a local clock-and-chronograph circuit at the other station. There were two repeaters on the main line, either at Albany and Buffalo or at New York and Buffalo. The signals were transmitted entirely through closed circuits. Care was always taken that the repeaters were kept in close adjustment during the exchange of signals.

The observations were made on May 13, 23, 24, 26, June 4 and 11, 1881, with Mr. Flint at Detroit and Mr. Wheeler at Cambridge; and on June 21, 22, 23, 24, and 29, 1881, with Mr. Flint at Cambridge and Mr. Wheeler at Detroit. The nights of May 13 and 23, being partially interrupted by clouds, were each considered as of half weight. On May 13, at Detroit, there was no reversal on a slow star, and the stars observed after signals were all with the clamp one way. On May 23, at Cambridge, the stars observed after signals were all with the clamp one way.

The chronographic readings were made in duplicate, and the reductions have been entirely in duplicate, a second person duplicating the work of the first. Collimation was determined from reversals on polar stars, except in two cases, where it was deduced by least squares from the observation-equations. The clock correction at a given epoch, the azimuth, and hourly rate, were determined by least squares from the observed times corrected for errors of level, reduction to middle wire, diurnal aberration, and usually for collimation. The observation-equations from which the most probable values of $\Delta\theta$, a, and ρ were deduced are of the form $Aa+R\rho+\Delta\theta+[t+Bb+C(c+\Delta i+ab'n)]-(\alpha+\text{ assumed }\Delta t)=v$, in which A, B, and C are the azimuth, level, and collimation factors respectively; R, the interval from a star observation to the given epoch, expressed in tenths of an hour; a, b, and c, the azimuth, level, and collimation corrections, respectively; ρ, the rate per hour, + when losing, − when gaining; Δt, the clock correction at the given epoch = assumed $\Delta t+\Delta\theta$; $\Delta\theta$, the correction to the assumed Δt; t, the clock-time of transit, usually the mean of 11 wires; Δi

Difference of Longitude, Detroit, Mich., and Cambridge, Mass.

TABLE 1.—TIME DETERMINATION.

[Detroit, Mich , May 13, 1881. A. R. Flint, Observer.]

Star.	Cl.	No. of wires.	b	Bb	$C(c+\Delta i+ab'n)$	Aa	$R\rho$	Clock time of transit $=t$	Right ascension $=\alpha$	$\alpha - t_c$
			s.	*s.*	*s.*	*s.*	*s.*	*h. m. s.*	*h. m. s.*	*m. s.*
θ Leonis	E.	11	+0.20	+0.19	−0.05	+0.20	0.00	11 09 27.55	11 08 02.30	−1 25.44
ξ Ursæ Maj. med	E.	11	+0.19	+0.22	−0.06	+0.09	0.00	11 13 18.02	11 11 52.66	−1 25.58
σ Leonis	E.	11	+0.23	+0.19	−0.05	+0.26	0.00	11 16 27.79	11 15 02.63	−1 25.35
ι Leonis	E.	11	+0.22	+0.19	−0.05	+0.23	0.00	11 19 11.08	11 17 45.84	−1 25.43
58 Ursæ Maj	E.	11	+0.19	+0.26	−0.07	−0.01	0.00	11 25 32.76	11 24 07.44	−1 25.58
υ Leonis	E.	11	+0.23	+0.17	−0.05	+0.30	0.00	11 32 19.20	11 30 54.05	−1 25.32
γ Cephei, L. C	E.	11	−0.08	+0.17	+0.34	+1.69	0.00	11 35 51.54	23 34 27.8	−1 23.9
χ Ursæ Maj	E.	11	+0.19	+0.28	−0.08	−0.07	0.00	11 41 14.23	11 39 48.78	−1 25.73
β Leonis	E.	11	+0.20	+0.18	−0.05	+0.21	0.00	11 44 27.34	11 43 02.10	−1 25.42
β Virginis	E.	11	+0.23	+0.17	−0.05	+0.28	0.00	11 45 57.72	11 44 32.56	−1 25.33
o Virginis	W.	11	+0.20	+0.17	+0.02	+0.24	0.00	12 00 36.91	11 59 11.67	−1 25.41
33 Bootis	W.	11	+0.08	+0.11	+0.03	−0.03	0.00	14 35 53.26	14 34 27.73	−1 25.64
μ Virginis	W.	11	+0.02	+0.01	+0.02	+0.32	0.00	14 38 15.91	14 36 50.80	−1 25.12
109 Virginis	W.	11	+0.03	+0.02	+0.02	+0.28	0.00	14 41 42.52	14 40 17.31	−1 25.23
47 Hev. Cephei, L. C	W.	8	−0.10	+0.26	−0.24	+1.95	0.00	14 51 42.10	2 50 18.8	−1 23.6
β Bootis	W.	11	+0.07	+0.09	+0.03	+0.01	0.00	14 58 56.22	14 57 31.01	−1 25.30
3 Serpentis	W.	11	+0.04	+0.03	+0.05	+0.26	0.00	15 10 45.02	15 09 19.84	−1 25.21
δ Bootis	W.	11	+0.08	+0.10	+0.02	+0.08	0.00	15 12 10.78	15 10 45.52	−1 25.36
μ Bootis	W.	11	+0.08	+0.10	+0.03	+0.04	0.00	15 21 28.25	15 20 02.83	−1 25.52
β Coronæ Borealis	W.	11	+0.08	+0.09	+0.02	+0.11	0.00	15 24 23.83	15 22 58.57	−1 25.35
ν^1 Bootis	W.	11	+0.08	+0.11	+0.03	+0.01	0.00	15 28 07.96	15 26 42.49	−1 25.58
ν^2 Bootis	W.	11	+0.08	+0.11	+0.03	+0.01	0.00	15 29 00.08	15 27 34.56	−1 25.63

Adopted rate $\rho = 0^s.00$.

OBSERVATION-EQUATIONS.

Epoch $13^h.0$ clock time. $\Delta t = -1^m\ 25^s.00 + \Delta\theta$.

					Weight.
+0.46a	+1.04c	$+\Delta\theta$	+0.45	$= v = +0.03$	1
+0.21	+1.18	+	+0.59	$= v = +0.05$	1
+0.59	+1.01	+	+0.36	$= v = \pm 0.00$	1
+0.53	+1.02	+	+0.44	$= v = +0.05$	1
−0.03	+1.39	+	+0.59	$= v = -0.06$	1
+0.68	+1.00	+	+0.33	$= v = +0.01$	1
+3.87	−4.43	+	−1.1	$= v = +0.3$	0.11
−0.16	+1.51	+	+0.75	$= v = +0.03$	1
+0.47	+1.04	+	+0.43	$= v = +0.02$	1
+0.64	+1.00	+	+0.34	$= v = \pm 0.00$	1
+0.55	−1.01	+	+0.37	$= v = +0.11$	1
−0.06a	−1.41c	$+\Delta\theta$	+0.58	$= v = +0.08$	1
+0.74	−1.00	+	+0.08	$= v = -0.10$	1
+0.64	−1.00	+	+0.19	$= v = -0.03$	1
+4.47	+5.21	+	−1.4	$= v = -0.3$	0.08
+0.03	−1.32	+	+0.25	$= v = -0.22$	1
+0.60	−1.01	+	+0.19	$= v = -0.05$	1
+0.18	−1.20	+	+0.31	$= v = -0.10$	1
+0.10	−1.26	+	+0.47	$= v = +0.03$	1
+0.25	−1.15	+	+0.30	$= v = -0.08$	1
+0.03	−1.33	+	+0.52	$= v = +0.05$	1
+0.02	−1.33	+	+0.58	$= v = +0.11$	1

NORMAL EQUATIONS.

$$+6.97a + 0.20c + 7.26\Delta\theta + 1.04 = 0$$
$$+0.20 + 31.81 - 2.90 + 0.24 = 0$$
$$+7.26 - 2.90 + 20.19 + 7.92 = 0$$

RESULTS.

$a = +0^s.437$
$c = -0^s.061$ cl. E.
$\Delta\theta = -0^s.560$

Difference of Longitude, Detroit, Mich., and Cambridge, Mass.—Continued.

TABLE 13.—TIME DETERMINATION.

[Cambridge, Mass., May 13, 1881. O. B. Wheeler, Observer.]

Star.	Cl.	No. of wires.	b	Bb	$C(c+\Delta i+ab'n)$	$A\alpha$	$R\rho$	Clock time of transit $=t$	Right ascension $=\alpha$	$\alpha - t_c$
			s.	s.	s.	s.	s.	h. m. s.	h. m. s.	s.
ν Leonis	W.	11	−0.04	−0.03	−0.25	+0.01	+0.03	11 30 54.87	11 30 54.05	−0.54
χ Ursæ Maj	W.	11	−0.04	−0.06	−0.38	0.00	+0.03	39 49.83	39 48.79	−0.60
β Leonis	W.	11	−0.04	−0.04	−0.26	0.00	+0.03	43 02.95	43 02.10	−0.55
Gr. 1852	W.	11	−0.04	−0.15	−1.18	−0.02	+0.02	59 18.27	59 16.02	−0.9
Br. 6, L. C	W.	11	−0.04	+0.08	+1.07	+0.03	+0.02	12 09 28.74	0 09 29.23	−0.7
20 Comæ	W.	11	−0.04	−0.04	−0.27	0.00	+0.02	23 48.21	12 23 47.50	−0.40
24 Comæ seq	W.	11	−0.04	−0.04	−0.27	0.00	+0.02	29 13.37	29 12.59	−0.47
γ Virginis med	W.	11	−0.04	−0.03	−0.25	+0.01	+0.02	35 41.53	35 40.84	−0.41
δ Virginis	E.	11	−0.01	−0.00	+0.22	+0.01	+0.02	49 39.86	49 39.58	−0.50
ϵ Virginis	E.	11	−0.01	−0.01	+0.23	0.00	+0.02	56 18.48	56 18.23	−0.47
θ Virginis	E.	11	−0.01	−0.00	+0.22	+0.01	+0.01	13 03 50.75	13 03 50.49	−0.46
α Ursæ Min., L. C	E.	4	+0.01	−0.37	−4.93	+0.29	+0.01	14 48.38	1 14 41.8	−1.3
α Ursæ Min., L. C	W.	7	−0.03	+0.74	+6.32	+0.29	+0.01	14 37.55	1 14 41.8	−2.8
τ Bootis	W.	11	−0.03	−0.03	−0.26	0.00	+0.01	41 40.20	13 41 39.57	−0.34
η Bootis	W.	11	−0.03	−0.03	−0.27	0.00	+0.01	49 04.86	49 04.27	−0.29
ϕ Bootis	W.	11	−0.01	−0.02	−0.33	0.00	−0.01	15 33 37.14	15 33 36.39	−0.40
ζ Coron. Bor. seq	W.	11	−0.01	−0.02	−0.32	0.00	−0.01	34 57.69	34 56.94	−0.42
α Serpentis	W.	11	−0.01	−0.01	−0.15	+0.01	−0.01	38 28.27	38 27.74	−0.37
β Serpentis	W.	11	−0.01	−0.01	−0.26	0.00	−0.01	40 45.72	40 44.97	−0.48
μ Serpentis	W.	11	−0.01	−0.01	−0.25	+0.01	−0.01	43 28.86	43 28.08	−0.52
ζ Ursæ Min	W.	5	−0.01	−0.05	−0.71	−0.03	−0.01	48 25.55	48 24.7	−0.1
ζ Ursæ Min	E.	6	+0.10	+0.41	+0.56	−0.03	−0.01	48 24.22	48 24.7	−0.5
ϕ Herculis	E.	11	+0.09	+0.12	+0.31	0.00	−0.01	16 05 04.64	16 05 04.21	−0.86
δ Ophiuchi	E.	11	+0.09	+0.06	+0.22	+0.01	−0.01	08 10.29	08 10.05	−0.52
ϵ Ophiuchi	E.	11	+0.09	+0.06	+0.22	+0.01	−0.01	12 05.20	12 05.01	−0.47
γ Herculis	E.	11	+0.09	+0.08	+0.23	0.00	−0.02	16 43.62	16 43.41	−0.52
ω Herculis	E.	11	+0.09	+0.08	+0.23	0.00	−0.02	19 58.89	19 58 64	−0.56
λ Ophiuchi	E.	11	+0.09	+0.06	+0.22	+0.01	−0.02	24 58.40	24 58.08	−0.60
σ Herculis	E.	11	+0.07	+0.09	+0.30	0.00	−0.02	30 19.21	30 19.07	−0.53
ζ Herculis	E.	11	+0.07	+0.08	+0.26	0.00	−0.02	36 51.23	36 51.11	−0.46
η Herculis	E.	11	+0.07	+0.08	+0.29	0.00	−0.02	38 52.24	38 52.05	−0.56
κ Ophiuchi	E.	11	+0.07	+0.05	+0.22	0.00	−0.02	52 05.62	52 05.35	−0.54
ϵ Ursæ Min	E.	11	+0.07	+0.37	+1.64	−0.04	−0.02	58 16.00	58 17.3	−0.7

Adopted collimation $=+0^s.130$ Clamp East. (See Table 12.)

OBSERVATION-EQUATIONS.

Epoch $14^h.6$ clock time. $\Delta t = +0^s.00 + \Delta\theta$.

	Weight.
$+0.68\alpha - 3.1\rho + \Delta\theta + 0.54 = v = +0.09$	1
$-0.16\ -2.9\ +\ +0.60 = v = +0.14$	0.65
$+0.47\ -2.9\ +\ +0.55 = v = +0.09$	1
$-2.67\ -2.6\ +\ +0.9\ = v = +0.4$	0.11
$+3.70\ -2.4\ +\ +0.7\ = v = +0.2$	0.13
$+0.38\ -2.2\ +\ +0.40 = v = -0.07$	1
$+0.42\ -2.1\ +\ +0.47 = v =\ 0.00$	1
$+0.68\ -2.0\ +\ +0.41 = v = -0\ 05$	1
$+0.62\ -1.8\ +\ +0.50 = v = +0.04$	1
$+0.52\ -1.7\ +\ +0.47 = v =\ 0.00$	1
$+0.74\ -1.5\ +\ +0.46 = v = -0.01$	1
$+32.60\ -1.4\ +\ +2.1\ = v = +1.9$	0.001
$+0.43\ -0.9\ +\ +0.34 = v = -0.14$	1
$+0.42\ -0.8\ +\ +0.29 = v = -0.19$	1
$+0.04\alpha + 1.0\rho + \Delta\theta + 0.40 = v = -0.10$	1

	Weight.
$+0.12\alpha + 1.0\rho + \Delta\theta + 0.42 = v = -0.08$	1
$+0.59\ +1.0\ +\ +0.37 = v = -0.12$	1
$+0.47\ +1.1\ +\ +0.48 = v = -0.02$	1
$+0.71\ +1.1\ +\ +0.52 = v = +0.03$	1
$-2.84\ +1.2\ +\ +0.3\ = v = -0.2$	0.1
$-0.07\ +1.5\ +\ +0.86 = v = +0.36$	0.7
$+0.72\ +1.5\ +\ +0.52 = v = +0.03$	1
$+0.73\ +1.6\ +\ +0.47 = v = -0.02$	1
$+0.42\ +1.7\ +\ +0.52 = v = +0.01$	1
$+0.48\ +1.7\ +\ +0.56 = v = +0.05$	1
$+0.64\ +1.8\ +\ +0.60 = v = +0.10$	1
$-0.01\ +1.9\ +\ +0.53 = v = +0.02$	1
$+0.21\ +2.0\ +\ +0.46 = v = -0.05$	1
$+0.07\ +2.0\ +\ +0.56 = v = +0.05$	1
$+0.55\ +2.3\ +\ +0.54 = v = +0.03$	1
$4.74\ +2.4\ +\ +0.7\ = v = +0.2$	0.04

NORMAL EQUATIONS.

$$+11.66\alpha - 2.62\rho + 10.69\Delta\theta + 5.20 = 0$$
$$-2.62 + 86.16 + 1.49 + 1.54 = 0$$
$$+10.69 + 1.49 + 25.73 + 12.62 = 0$$

RESULTS.

$\Delta\theta = -0^s.493$

$\rho = -0^s.009$

$\alpha = +0^s.009$

Difference of Longitude, Detroit, Mich., and Cambridge, Mass.—Continued.

TABLE 26.—FINAL RESULTS.

O. B. WHEELER AT CAMBRIDGE. A. R. FLINT AT DETROIT.

Date.	Difference of time.			Wt.	Correction.	Corrected results.	Double wave-time, east *minus* west.
	Signals from Cambridge.	Signals from Detroit.	Mean.				
1881.	*h. m. s.*	*h. m. s.*	*h. m. s.*		*s.*	*h. m. s.*	*s.*
May 13	0 47 41.064	0 47 41.262	0 47 41.163	0.5	+0.034	0 47 41.197	+0.198
23	40.831	41.099	40.966	0.5	+0.034	41.010	+0.268
24	40.932	41.144	41.038	1	+0.034	41.072	+0.212
26	40.907	41.154	41.030	1	+0.034	41.064	+0.247
June 4	40.948	41.220	41.084	1	+0.034	41.118	+0.272
11	40.848	41.177	41.012	1	+0.034	41.046	+0.329
Weighted mean			0 47 41.046				

A. R. FLINT AT CAMBRIDGE. O. B. WHEELER AT DETROIT.

June 21	0 47 41.022	0 47 41.286	0 47 41.154	1	−0.034	0 47 41.120	+0.264
22	41.050	41.292	41.171	1	−0.034	41.137	+0.242
23	41.010	41.267	41.138	1	−0.034	41.104	+0.257
24	40.973	41.246	41.110	1	−0.034	41.076	+0.273
29	40.857	41.134	40.995	1	−0.034	40.961	+0.277
Weighted mean			0 47 41.114				
Difference of weighted means = personal and instrumental equation			= 0.068				
Correction = half of difference			= 0.034				

	h.	*m.*	*s.*
Weighted mean of corrected results	0	47	41.080
Reduction to east pier of Detroit Observatory	−		0.004
East pier of Lake-Survey Observatory, Detroit, west of Coast-Survey post, Cambridge	0	47	41.076
Longitude of Coast-Survey post, Cambridge, Appendix 6, Coast-Survey Report of 1880	4	44	31.090
Longitude of east pier of Lake-Survey Observatory at Detroit west of Greenwich	5	32	12.166